ModelIt!

Teacher's Guide

Can You Hear a Tsunami Coming?

How Early Detection Systems Save Lives from Giant Waves

Quick Reference

Grade Level	Grade 04L2
NGSS Standard	4-PS4-3, 4-ESS3-2
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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Table of Contents

ModelIt!

1 NGSS Standards Unpacking & CAST Alignment 3

2 Scoring Rubric 4

3 Background Content Knowledge 5

4 The LEVER Framework 6

5 Slide-by-Slide Facilitation Guide 7

6 Answer Key & Expected Responses 9

7 Collaborative Learning Protocols 10

8 Differentiation & Extensions 11

9 Common Misconceptions 12

1 NGSS Standards & CAST Alignment

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Performance Expectation: 4-PS4-3, 4-ESS3-2

Generate and compare multiple solutions that use patterns to transfer information. Generate and compare multiple solutions to reduce the impacts of natural Earth processes on humans.

Three-Dimensional Learning

Science Practice	**Developing and Using Models / Designing Solutions** — Students develop a five-component model of tsunami detection and warning, then design solutions to maximize evacuation success given constraints of distance and wave energy.
Disciplinary Core Idea	**PS4.C Information Technologies and Instrumentation / ESS3.B Natural Hazards** — Patterns in waves can be used to transfer information through detection systems. Natural hazards like tsunamis can be reduced through engineering solutions and warning systems.
Crosscutting Concept	**Cause and Effect** — Students identify causal chains from earthquake to wave energy to detection to warning to evacuation, understanding how each link affects the final outcome of human safety.

CAST Testing Alignment

MULTIPLE CHOICE:

Two coastal towns experience the same tsunami. Town A is 400 miles from the earthquake and has a warning siren system. Town B is 40 miles from the earthquake and has the same system. Town A evacuates successfully but Town B has many casualties. Which best explains the difference?

CONSTRUCTED RESPONSE:

A new coastal town is being built in a tsunami-risk area. Using what you know about wave energy, detection, and warning systems, write a plan for keeping residents safe. Use the words 'detection time,' 'warning speed,' and 'evacuation success' in your answer.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

Tsunamis are among the most destructive natural disasters on Earth, but they are also among the most predictable and survivable — IF communities have proper warning systems. The 2004 Indian Ocean tsunami killed over 230,000 people partly because there was no warning system in place. Since then, the world has invested billions in detection technology and warning networks that have saved countless lives.

How Tsunamis Form

Tsunamis begin when a massive disturbance displaces ocean water — usually an underwater earthquake, but also volcanic eruptions or landslides. The displaced water creates waves that radiate outward in all directions, similar to ripples from dropping a stone in a pond but on an enormous scale. In deep ocean, tsunami waves travel at speeds up to 500 mph (as fast as a jet airplane!) but are only about 1 to 3 feet tall, making them nearly invisible. As the waves approach shallow coastal water, they slow down and compress, growing into walls of water 30 to 100 feet tall.

How Detection Works

The Pacific Tsunami Warning System uses a network of ocean-floor sensors called DART buoys (Deep-ocean Assessment and Reporting of Tsunamis). These sensors sit on the ocean floor and measure water pressure. When a tsunami wave passes overhead, the increased water pressure triggers the sensor, which sends data to a surface buoy via acoustic signal. The surface buoy transmits the data to a satellite, which relays it to warning centers. Seismographs also detect the earthquakes that cause tsunamis, providing the first alert within minutes.

The Race Against Time

After detection, the warning must reach people fast enough for them to evacuate. Modern systems use multiple channels: outdoor sirens, emergency phone alerts, radio and TV broadcasts, and even social media. The challenge is that tsunamis from nearby earthquakes may arrive in as little as 10 to 30 minutes, while warnings take time to generate and distribute. This is why coastal communities conduct regular evacuation drills — when a real warning comes, everyone must already know where to go and how to get there without hesitation.

Engineering Solutions for Close Tsunamis

For tsunamis that originate close to shore, traditional warning systems may not provide enough time. Engineers have developed several solutions: vertical evacuation structures (tall, reinforced buildings or towers that people can climb above the wave height), earthquake-triggered automatic sirens (that sound immediately when a large earthquake is detected without waiting for wave confirmation), tsunami walls and barriers (sea walls designed to absorb or redirect wave energy), and natural barriers (mangrove forests and coral reefs that slow wave energy). Japan, which experiences frequent tsunamis, has invested heavily in all of these approaches.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L **Locate the System**

Students identify Wave Energy and Distance from Shore as external components (natural conditions the system cannot control) and Detection Time, Warning Speed, and Evacuation Success as internal components that respond to the external conditions and to each other.

E **Establish Relationships**

Students determine that Wave Energy positively affects Detection Time (bigger waves are easier to detect), Distance from Shore positively affects Warning Speed (more distance means more time to communicate), Warning Speed positively affects Evacuation Success, Wave Energy negatively affects Evacuation Success (more powerful waves are harder to survive), and Detection Time positively affects Warning Speed.

V **Visualize & Model**

Students build a five-component model showing how earthquake energy creates detectable waves, distance provides warning time, and the chain from detection to warning to evacuation determines whether people survive.

E **Evaluate Outcomes**

Students run four scenarios (far/strong, close/strong, far/weak, close/weak) to observe how the combination of wave energy and distance determines whether a warning system can save people.

R **Revise & Extend**

Students add coastal shape, population density, or evacuation practice to explore why some communities are better prepared than others and why engineering solutions are needed for close-source tsunamis.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

2 min

Say: *"Imagine you live near the ocean and an earthquake just happened far away. A giant wave is coming. How would you even KNOW it is coming?"*

Watch For

Show a brief video clip of a tsunami warning siren sounding. Ask: What would you do if you heard this? How much time do you think you would have?

Slide 2: Learning Objectives

3 min

Say: *"Today we are modeling something that saves real lives every year — tsunami warning systems. We are going to figure out what makes the difference between survival and disaster."*

Watch For

Have students read objectives aloud. Pre-teach 'detection' and 'evacuation' with simple examples (smoke detector detects fire, fire drill practices evacuation).

Slide 3: Big Question

3 min

Say: *"The earthquake happens HERE, under the ocean. The town is HERE, on the coast. The wave is coming. Can the warning system alert people in time?"*

Watch For

Turn and talk: What do you think matters more — how strong the tsunami is or how far away it starts? Write predictions.

Slide 4: LEVER Framework

2 min

Say: *"We have two things nature controls — how strong the wave is and how far away it starts. Then three things that determine if people survive."*

Watch For

Preview all five components. Ask: Which component do you think is most important for saving lives?

Slide 5: Activity 1: Components

8 min

Say: *"Wave Energy and Distance from Shore are external — nature decides these. Detection Time, Warning Speed, and Evacuation Success are the internal responses."*

Watch For

Guide component sorting. Discuss why we cannot control earthquakes or distance, but we CAN engineer better detection, faster warnings, and practiced evacuations.

Slide 6: Activity 2: Connections

10 min

Say: *"Here is the tricky part: Wave Energy has BOTH a positive and negative effect. It makes detection EASIER but also makes survival HARDER. Distance helps everything."*

Watch For

Map relationships. Emphasize the competing effects of wave energy — easier to detect but harder to survive.

Slide 7: Activity 3: Simulation

12 min

Say: *"Watch what happens to evacuation success in each scenario. The close, powerful tsunami is terrifying — look at how little time people have."*

Watch For

Run all four scenarios. Focus on comparing far/strong vs. close/strong — same wave energy, dramatically different outcomes.

Slide 8: Discoveries

5 min

Say: *"Distance is the biggest gift a warning system has. More distance means more time, and more time means more lives saved."*

Watch For

Create a class timeline: earthquake happens (0 min), sensors detect (2 min), warning sent (5 min), sirens sound (7 min), evacuation (7-60 min). How much time does each scenario give?

Slide 9: STEM Challenge

12 min

Say: *"Design the ultimate tsunami warning system for a town that faces BOTH far-away and nearby tsunami threats. How do you save lives when you might only have 10 minutes?"*

Watch For

Groups design warning systems on poster boards — include sensor placement, communication methods, evacuation routes, and vertical evacuation structures for close tsunamis.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Wave Energy, Distance from Shore
- **INTERNAL:** Detection Time, Warning Speed, Evacuation Success

Relationships

- **Wave Energy to Detection Time:** POSITIVE (+) — Bigger, more energetic tsunami waves create stronger pressure signals on ocean-floor sensors, making them easier and faster to detect. A magnitude 9.0 earthquake creates unmistakable signals that sensors pick up almost instantly, while a smaller event might take longer to confirm.
- **Distance from Shore to Warning Speed:** POSITIVE (+) — Greater distance between the tsunami's origin and the coast provides more time for the entire warning chain — detection, analysis, alert generation, and communication to the public. A tsunami 1,000 miles away gives hours of warning; one 50 miles away gives minutes.
- **Warning Speed to Evacuation Success:** POSITIVE (+) — Faster warnings give people more time to reach safe ground. Every additional minute of warning time means more people can evacuate. Studies show that each 10-minute increase in warning time can increase survival rates by significant percentages.
- **Wave Energy to Evacuation Success:** NEGATIVE (-) — More powerful tsunamis are harder to survive even with warning. Higher waves flood larger areas, reach farther inland, and are more destructive. People need to evacuate farther and higher to escape a powerful tsunami compared to a weaker one.
- **Detection Time to Warning Speed:** POSITIVE (+) — Faster detection feeds directly into faster warnings. If sensors detect the tsunami within 2 minutes instead of 10 minutes, the warning system gains 8 additional minutes to alert the public and begin evacuations.

Key Discoveries

- Greater distance from shore gives warning systems MORE time to detect the tsunami and alert people, which increases evacuation success
- Higher wave energy makes the tsunami easier to detect (bigger signals for sensors) but also more destructive, which works AGAINST evacuation success
- The most dangerous scenario is high wave energy with short distance — the waves arrive before people can evacuate
- Warning speed is critical: even with early detection, slow communication means people do not get the message in time to reach safety

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

ModelIt!

Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Tsunamis are just really big waves like surfing waves"	Tsunamis are fundamentally different from wind-generated ocean waves. Regular waves are created by wind at the surface and only move the top layer of water. Tsunamis are caused by the displacement of the ENTIRE water column from the ocean floor to the surface. A regular wave might contain thousands of gallons of water. A tsunami contains BILLIONS of gallons pushing forward as a massive wall of water that does not just crash on shore — it floods inland for miles, carrying debris, and then rushes back out with equal force.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"If you can see the tsunami wave coming, you still have time to run"	By the time you can see a tsunami wave approaching the shore, it is usually too late to escape on foot. Tsunami waves travel at 20 to 30 mph near shore — faster than any person can run. The wave may appear as a wall of water 30 to 100 feet tall moving faster than a car in a school zone. Additionally, the first sign of a tsunami is often the water pulling BACK from shore (exposing the sea floor), which happens just minutes before the wave arrives. That is why warning systems and pre-planned evacuations are essential.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Tsunamis only happen in the Pacific Ocean"	While the Pacific Ocean has the most tsunami activity due to the Ring of Fire (a zone of frequent earthquakes), tsunamis can occur in ANY body of water. The 2004 Indian Ocean tsunami killed over 230,000 people. The Mediterranean Sea, Caribbean Sea, and even large lakes have experienced tsunamis. Any body of water near a fault line, volcano, or steep underwater slope can generate a tsunami.